

# The Dimension of Nonterminating Computations

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## Abstract

How much randomness can remain in a random tape on which a computation runs forever? For a finite variable-model resampling computation, reversing a finite run yields an exact likelihood identity: the self-information of the repaired events minus the code length of the repair sequence equals the log-likelihood ratio between a backward record law and the source law of the random choices used by the run. If this difference is  $\Delta_T$ , then

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}.$$

A linear lower bound on  $\Delta_T$  for every execution lasting  $T$  repairs therefore gives an exponential running-time tail and, under computability, an upper bound on the Kolmogorov complexity of the random data used by each such execution. For rejection sampling from a computable finite source  $\pi$  with rejection set  $U$ , let  $\delta \in [0, 1]$  satisfy  $\sum_{x \in U} \pi(x)^\delta = 1$ . Every tape rejected forever has effective strong source-relative dimension at most  $\delta$ , the bound is attained, and the nonterminating set has source-relative Hausdorff dimension  $\delta$ . Finally, two whole-block resampling systems are constructed with the same dependency graph and event marginals but different survival exponents, different maximal effective strong dimensions, and different Hausdorff dimensions. Thus the graph and marginals determine none of these three quantities, even when every repair redraws the entire random block.

## 1 Introduction

A randomized algorithm maps its random tape into a trajectory. Once the tape is fixed, the computation is deterministic: it either stops or it does not. Let

$$\mathcal{N} = \{\omega : \tau(\omega) = \infty\},$$

where  $\tau$  is the number of rejection or repair steps before stopping and  $\tau = \infty$  on a nonterminating tape. The tail probability  $\mathbb{P}\{\tau \geq n\}$  measures how often the computation survives  $n$  steps. When the limit exists, define the *survival exponent*

$$\alpha = - \lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}.$$

The question here is pointwise: how incompressible can a single tape in  $\mathcal{N}$  be?

The paper makes two contributions. First, an exact backward likelihood identity for finite variable-model resampling computations yields both running-time tail bounds and Kolmogorov-complexity bounds for individual surviving tape prefixes. Second, a nested system and a disjoint system with the same dependency graph and event marginals need not determine the survival exponent or either dimension quantity. Before the general identity is developed, rejection sampling serves as the basic case: its nonterminating tapes form a self-similar set whose pointwise and Hausdorff dimensions can be computed explicitly.

## 1.1 Rejection sampling: the basic case

A familiar example is the simplest form of rejection sampling. Draw  $X_1, X_2, \dots$  independently from a full-support probability law  $\pi$  on a finite set  $\Omega$  with  $|\Omega| \geq 2$ , reject a draw in  $U \subseteq \Omega$ , and return the first draw in  $G = \Omega \setminus U$ . This is the special case of the classical accept–reject method in which each proposed value is accepted or rejected deterministically (von Neumann, 1951; Devroye, 1986).

Write  $p = \pi(U)$  and  $a = \pi(G) = 1 - p$ . If  $0 < a < 1$ , then

$$\mathbb{P}\{\tau \geq n\} = p^n, \quad \mathbb{P}\{\tau = t\} = p^t a, \quad \mathbb{E}\tau = \frac{p}{a}, \quad \mathbb{E}(\tau + 1) = \frac{1}{a} \quad (n, t \in \mathbb{Z}_{\geq 0}). \quad (1)$$

Here  $\tau$  is the number of rejected draws and  $\tau + 1$  is the total number of draws. For  $a > 0$ , the returned value has the conditional law

$$\mathbb{P}\{X_{\tau+1} \in B\} = \frac{\pi(B \cap G)}{\pi(G)} \quad (B \subseteq \Omega). \quad (2)$$

If  $a = 1$ , the first draw is accepted; if  $a = 0$ , the algorithm never stops. When  $0 < a < 1$ , the algorithm stops almost surely although the set  $U^{\mathbb{N}}$  of tapes on which it never stops is nonempty. Probability assigns this set mass zero. Dimension asks how much randomness can nevertheless remain in one of its elements.

Now suppose that the source is uniform and  $U \neq \emptyset$ . Write  $Q = |\Omega| \geq 2$  and  $M = |U|$ , so  $1 \leq M \leq Q$ . If  $E_n$  is the event that the first  $n$  blocks are rejected, then

$$\mathbb{P}(E_n) = \left(\frac{M}{Q}\right)^n,$$

and there are only  $M^n$  possible surviving prefixes. Each such prefix can be described using  $n \log_2 M + o(n)$  bits, whereas the source supplies  $n \log_2 Q$  bits. Their ratio is

$$\frac{\log_2 M}{\log_2 Q}.$$

In the two-bit example

$$\Omega = \{00, 01, 10, 11\}, \quad U = \{00, 01\}.$$

Every surviving block has its first bit fixed and its second bit free. Exactly one half of the source information can remain.

For a computable full-support source  $P = \pi^{\mathbb{N}}$ , let  $\omega_{1:n}$  denote the first  $n$  source symbols (or blocks), and define their self-information by

$$I_P(\omega_{1:n}) = -\log_2 P[\omega_{1:n}].$$

Let  $K(w)$  be the length of the shortest program for  $w$  on a fixed universal prefix-free machine. The effective strong  $P$ -dimension of a tape  $\omega$  is

$$\text{Dim}_{\text{eff}, P}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{I_P(\omega_{1:n})}. \quad (3)$$

Replacing the limsup by a liminf defines the effective  $P$ -dimension  $\text{dim}_{\text{eff}, P}(\omega)$ . For a fair binary source, write

$$\text{Dim}_{\text{eff}}(\omega) = \limsup_{n \rightarrow \infty} \frac{K(\omega_{1:n})}{n}.$$

This is the constructive strong dimension relative to a source measure (Athreya et al., 2007; Lutz, 2011). It is recalled here for rejection sampling; later sections relate it to nontermination in general resampling computations. A value near one means that the tape is nearly as incompressible as an ordinary draw from  $P$ . A smaller value means that continued execution has imposed structure on the tape.

For a set  $E$  of tapes,  $\dim_H^P E$  denotes its Billingsley, or source-relative Hausdorff, dimension: in the Hausdorff covering construction, the cylinder  $[w]$  is assigned diameter  $P[w]$  (Billingsley, 1965; Lutz, 2011). Under a uniform source, this is ordinary Hausdorff dimension in the usual symbolic metric.

There are three different questions:

- (i)  $\mathbb{P}(E_n)$  is the probability of running for at least  $n$  steps;
- (ii)  $\text{Dim}_{\text{eff},P}(\omega)$  measures the randomness of one tape; and
- (iii)  $\dim_H^P \mathcal{N}$  measures the size of the set of all nonterminating tapes.

For rejection sampling, the tail in (i) is  $\pi(U)^n$ . The two dimension questions are governed by the number  $\delta$  that solves

$$\sum_{x \in U} \pi(x)^\delta = 1. \quad (4)$$

Then every tape rejected forever satisfies

$$\text{Dim}_{\text{eff},P}(\omega) \leq \delta \quad (\omega \in U^\mathbb{N}). \quad (5)$$

Moreover,

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}.$$

This is a probability law, and a tape that is Martin–Löf random under  $Q_\delta^\mathbb{N}$ —hence algorithmically typical for that law—attains equality. The classical set calculation gives  $\dim_H^P(U^\mathbb{N}) = \delta$ . Under the uniform law,  $\pi(U) = Q^{\delta-1}$ , so the tail exponent and the dimension determine one another. For a nonuniform law, they need not.

Effective dimension is pointwise, whereas Hausdorff dimension measures the entire nonterminating set. The law  $Q_\delta$  produces tapes attaining the pointwise bound; the source-relative Hausdorff dimension of the whole set equals the same value  $\delta$ .

## 1.2 Reconstructing the random tape

Rejection sampling is the simplest resampling computation. The same question can be posed for finite variable-model resampling algorithms. At each step, such an algorithm selects a bad event and redraws the variables on which that event depends. A finite run can be read backward from its final assignment, its ordered list of repairs, and the values erased by those repairs. Restoring the erased values in reverse order recovers every random choice used by the run.

This reconstruction gives a simple comparison:

- information revealed by the repaired events
- information needed to describe the repair history = a log-likelihood ratio.

Because the backward descriptions have total probability at most one, this ratio has an exponential upper tail. If the description is computable, the random choices used by the run admit a description shorter than their source self-information by at least the same number of bits. Theorem 2.1 formalizes both conclusions. Event probabilities alone are not enough; the description must include the repair order and any control information needed to reverse the run.

### 1.3 Nested rejection rules

A dependency graph has one vertex for each rejection rule and joins two rules that may interact. In the variable model, two rules are joined when they depend on a common random variable. The Lovász local lemma and its variants use this graph to give conditions under which all bad events can be avoided (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005). Algorithmic analyses use the same graph to bound the number and weight of repair histories (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2022). The graph does not record how the rejected sets intersect or whether one contains another.

Suppose several nested rules can reject the same random block. The algorithm records the smallest rule that applies and then redraws the whole block. Its true one-step survival probability is

$$\pi \left( \bigcup_A A \right),$$

whereas the complete-graph history enumeration considered below assigns the one-step weight  $\sum_A \pi(A)$ , counting an outcome once for every event that contains it. Dividing the union into disjoint cases preserves the recorded rule and recovers the exact survival probability. Section 3 gives nested Hamming balls for which the graph enumeration yields only the trivial dimension bound, followed by two systems with the same graph and marginals but different survival exponents and dimensions.

Because every rejection redraws the whole block, this example has no persistent state. Partially overlapping local repairs, which retain variables relevant to later repairs, are the harder case.

### 1.4 Contribution and limits

The paper has two main contributions. The first is Theorem 2.1, an exact likelihood-ratio identity for every finite run of a finite variable-model resampling computation. A probability bound applied to this identity controls the running time; a code built from the same backward law controls the Kolmogorov complexity of each surviving tape prefix. These are two uses of one equality, rather than separate information-theoretic and computational arguments.

The second contribution is an explicit separation. Theorem 3.1 computes the survival exponent and dimension from the union of the rejected sets. Example 3.5 gives two whole-block resampling computations with the same dependency graph and the same vector of bad-event probabilities but different survival exponents, different maximal effective strong dimensions of individual nonterminating tapes, and different Hausdorff dimensions of their nonterminating sets. The author is not aware of an earlier pair with identical graph and marginal data that separates all three quantities. This is not a claim that the loss of intersection information by dependency graphs is itself new: intersection-sensitive improvements over graph-based Shearer bounds are known (Shearer, 1985; He et al., 2017, 2023).

The present analysis does not yet give matching upper and lower bounds for partially overlapping local repairs with persistent state. When one repair resamples only part of the variables involved in another, the retained values may affect the next repair. The repair sequence alone may therefore be insufficient to predict or reconstruct the run. The open problem is to find a smaller state description that still determines the survival exponent and permits backward reconstruction.

### 1.5 Related work

The Lovász local lemma asks whether all bad events can be avoided; its algorithmic forms ask whether resampling can find such an outcome efficiently (Erdős and Lovász, 1975; Moser and Tardos,

2010). The present work asks a different quantitative question: how quickly long runs become rare, and how large and complex the exceptional set of nonterminating tapes is.

The analysis builds on two established lines of work. For finite-state systems, the relation among weighted path growth, Hausdorff dimension, and the Markov law obtained by reweighting paths is classical (Parry, 1964; Mauldin and Williams, 1988; Bowen, 1975; Simpson, 2015). Entropy-compression arguments encode repair histories reversibly, and their interpretation through Kolmogorov complexity is also known (Moser, 2009; Moser and Tardos, 2010; Fortnow, 2009; Tao, 2009). These results are used as tools rather than claimed as new.

Recent work also assigns pointwise complexity to a realized neural network or to a location in a Gaussian field (Li and Xu, 2026; Xu, 2026b). Here the realized object is a random tape. A related representation question asks how much state information is needed for a Bellman recursion (Xu, 2026a); in the present setting, that question reappears for overlapping local repairs. These connections are motivational only: the results below do not directly yield learning-theoretic generalization bounds. Nevertheless, the study of nonterminating computations may contribute to deriving sharp, computable bounds for deep learning and field theory; see Section 2.2 of Li and Xu (2026) and Appendix B of Xu (2026b).

## 2 Reversing a resampling computation

Let  $X_1, \dots, X_n$  be independent finite-valued variables with product law  $\pi = \bigotimes_i \pi_i$ . Each nonempty bad event  $A$  is determined by the coordinates  $v(A)$  and has probability  $p_A > 0$ . Fix a deterministic rule for selecting a currently true bad event.

List the random choices in the order in which the algorithm uses them. First draw  $S_0 \sim \pi$ . Whenever the current assignment contains a bad event, select  $A_t$ , draw  $Z_t \sim \pi_{v(A_t)}$  independently of the past, and write  $Z_t$  on  $v(A_t)$ . Let  $E_T$  be the event that at least  $T$  resamplings occur, and put

$$U_T = (S_0, Z_1, \dots, Z_T).$$

Thus  $E_T = \{\tau \geq T\}$ . For each fixed  $T$ , the minimal tape prefixes that realize  $T$  resamplings form a prefix-free stopping line. Their cylinders are disjoint, and their probabilities sum to  $\mathbb{P}(E_T)$ .

Let  $Y_t = S_{t-1}|_{v(A_t)}$  be the configuration overwritten at time  $t$ , let  $S_T$  be the assignment after  $T$  repairs, and let  $L_T = (A_1, \dots, A_T)$ . Coordinatewise, the entries of the used tape are partitioned into the final assignment and the overwritten configurations. Therefore

$$P_{\text{tape}}(U_T) = \pi(S_T) \prod_{t=1}^T p_{A_t} \pi_{v(A_t)}(Y_t \mid A_t). \quad (6)$$

Let  $Q_T$  be a subprobability mass function on lists  $\ell = (A_1, \dots, A_T)$ , positive on every feasible  $T$ -repair list. Define a comparison subprobability law on complete records by

$$G_T(s, \ell, y_{1:T}) = \pi(s) Q_T(\ell) \prod_{t=1}^T \pi_{v(A_t)}(y_t \mid A_t). \quad (7)$$

The map

$$U_T \longmapsto (S_T, L_T, Y_{1:T})$$

is injective. Indeed, reverse the repairs from  $T$  to 1: the current values on  $v(A_t)$  recover  $Z_t$ , and restoring  $Y_t$  recovers the preceding assignment.

**Theorem 2.1** (Backward likelihood identity). *Every execution in  $E_T$  satisfies*

$$\log_2 \frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(U_T)} = \sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} - \log_2 \frac{1}{Q_T(L_T)}. \quad (8)$$

*Call the right side  $\Delta_T$  on  $E_T$  and set  $\Delta_T = -\infty$  otherwise. Then, for every  $u \geq 0$ ,*

$$\mathbb{P}\{\Delta_T \geq u\} \leq 2^{-u}. \quad (9)$$

*If the instance and the family  $(Q_T)_{T \geq 1}$  are uniformly computable, then*

$$-\log_2 P_{\text{tape}}(U_T) - K(U_T \mid T, \text{instance}) \geq \Delta_T - O(1). \quad (10)$$

*Proof.* Dividing (7) by (6) cancels the final-state factor and all conditional overwritten-value factors. The remaining terms give

$$\frac{G_T(S_T, L_T, Y_{1:T})}{P_{\text{tape}}(U_T)} = \frac{Q_T(L_T)}{\prod_{t=1}^T p_{A_t}} = 2^{\Delta_T},$$

which proves (8).

It remains to use this likelihood ratio. Let  $\mathcal{W}_T$  be the prefix-free set of minimal tape prefixes that produce at least  $T$  repairs, and let  $\mathcal{R}_T$  be the set of formal records on which  $G_T$  is defined. For  $w \in \mathcal{W}_T$ , write

$$R_T(w) = (S_T(w), L_T(w), Y_{1:T}(w)).$$

The reconstruction argument shows that  $R_T$  is injective.

First observe that  $G_T$  has total mass at most one. Indeed, for a fixed list  $\ell = (A_1, \dots, A_T)$ , the final-state law and each conditional overwritten-value law have total mass one. Consequently,

$$\sum_{r \in \mathcal{R}_T} G_T(r) = \sum_{\ell} Q_T(\ell) \leq 1.$$

Now pull  $G_T$  back to live tape prefixes by defining

$$\mu_T(w) = G_T(R_T(w)), \quad w \in \mathcal{W}_T.$$

Injectivity of  $R_T$  gives

$$\sum_{w \in \mathcal{W}_T} \mu_T(w) = \sum_{r \in R_T(\mathcal{W}_T)} G_T(r) \leq \sum_{r \in \mathcal{R}_T} G_T(r) \leq 1.$$

Thus  $\mu_T$  is a subprobability mass function. Its total mass can be strictly less than one because  $Q_T$  may itself have mass less than one, or because  $G_T$  may assign mass to formal records that are not produced by the algorithm.

For every  $w \in \mathcal{W}_T$ , the identity already proved gives

$$2^{\Delta_T(w)} = \frac{G_T(R_T(w))}{P_{\text{tape}}(w)} = \frac{\mu_T(w)}{P_{\text{tape}}(w)}.$$

Because  $\Delta_T = -\infty$  outside  $E_T$  and the cylinders determined by  $\mathcal{W}_T$  are disjoint, for every  $u \geq 0$ ,

$$\begin{aligned} \mathbb{P}\{\Delta_T \geq u\} &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} P_{\text{tape}}(w) \\ &= \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} 2^{-\Delta_T(w)} \mu_T(w) \\ &\leq 2^{-u} \sum_{\substack{w \in \mathcal{W}_T \\ \Delta_T(w) \geq u}} \mu_T(w) \\ &\leq 2^{-u}. \end{aligned}$$

This proves (9); equivalently, it is Markov's inequality applied to a subprobability likelihood ratio.

Finally, uniform computability of the instance and of  $(Q_T)_{T \geq 1}$  makes  $(\mu_T)_{T \geq 1}$  a uniformly computable family of subprobability mass functions. Shannon–Fano coding, or equivalently the coding theorem for computable subprobability masses (Shannon, 1948; Fano, 1949), therefore gives

$$K(U_T \mid T, \text{instance}) \leq -\log_2 \mu_T(U_T) + O(1).$$

Since

$$-\log_2 \mu_T(U_T) = -\log_2 P_{\text{tape}}(U_T) - \Delta_T,$$

rearranging proves (10).  $\square$

The two terms in (8) have different origins. The first is the information revealed by the events that were repaired. The second is the number of bits needed to describe which repairs occurred. Neither term alone gives a bound; their difference does, because it is a likelihood ratio.

For example, color the vertices of a graph independently with  $q$  colors and repeatedly recolor the endpoints of the first monochromatic edge. A fixed edge is monochromatic with probability  $1/q$ , so each repair contributes  $\log_2 q$  bits to the first term. The second term pays for the repaired edges and their order. The comparison is therefore between information ruled out by the violations and information used by the algorithm's history.

**Corollary 2.2** (A tail bound and a pointwise dimension bound). *Suppose every  $T$ -repair execution satisfies*

$$\sum_{t=1}^T \log_2 \frac{1}{p_{A_t}} \geq \beta T - C_0, \quad -\log_2 Q_T(L_T) \leq \gamma T + C_1,$$

where  $\beta > \gamma$ . Then

$$\mathbb{P}(E_T) \leq 2^{C_0 + C_1 - (\beta - \gamma)T}. \quad (11)$$

For the pointwise conclusion, assume in addition that

- (i) the instance and the family  $(Q_T)_{T \geq 1}$  are uniformly computable;
- (ii) the computation reads a fixed fair binary tape sequentially, the initial draw consumes  $c_0 = O(1)$  bits, and every repair consumes exactly  $c$  bits, so that  $U_T$  is uniformly computably interconvertible with  $\omega_{1:c_0+cT}$ .

Then every nonterminating tape satisfies

$$\text{Dim}_{\text{eff}}(\omega) \leq \max \left\{ 0, 1 - \frac{\beta - \gamma}{c} \right\}. \quad (12)$$

*Proof.* The assumptions give  $\Delta_T \geq (\beta - \gamma)T - (C_0 + C_1)$ . Equation (11) follows from (9). Equation (10), together with the  $O(\log_2 T)$  bits needed to specify  $T$ , gives  $K(U_T) \leq cT - (\beta - \gamma)T + o(T)$  along an infinite execution. Divide by  $cT$ ; the gap between consecutive repair prefixes is the fixed number  $c$  of bits.  $\square$

*Remark 2.3* (What the record must contain). For a recursive Moser-style implementation,  $L_T$  must be enlarged to include the recursion tree or return structure unless those data are recoverable by a separate prefix-free decoding rule. In Theorem 2.1, everything needed for reversal contributes to the probability  $Q_T(L_T)$ .

### 3 What a dependency graph forgets

To use Theorem 2.1, one must describe the possible lists of repairs. The Lovász local lemma summarizes event interactions by a dependency graph, and its algorithmic proofs use that graph to organize repair histories.

#### 3.1 Why dependency graphs enter

Let  $\mathcal{A}$  be a finite family of bad events. The Lovász local lemma asks when there is a positive probability of avoiding all of them:

$$\mathbb{P}\left(\bigcap_{A \in \mathcal{A}} A^c\right) > 0. \quad (13)$$

In the variable model, where independent variables generate the events, the standard graph  $D$  joins  $A$  and  $B$  whenever their variable sets overlap. For example, in random vertex coloring a bad event says that one edge is monochromatic, and two such events are adjacent in  $D$  when their edges share a vertex. More generally, write  $\Gamma(A)$  for the neighbors of  $A$ . A loopless graph on  $\mathcal{A}$  is a dependency graph if every event  $A$  is independent of  $\sigma(B : B \in \mathcal{A} \setminus (\{A\} \cup \Gamma(A)))$ . The vertices of  $D$  are events; this is different from the bipartite variable–event incidence graph used in Section 3.5. If  $\mathbb{P}(A) \leq p$  for every  $A$  and  $D$  has maximum degree  $d$ , the symmetric local lemma gives the familiar sufficient condition

$$ep(d+1) \leq 1. \quad (14)$$

More generally, Shearer’s region gives the sharp worst-case criterion based only on a graph and the marginal event probabilities (Erdős and Lovász, 1975; Shearer, 1985; Scott and Sokal, 2005).

Moser and Tardos turn this existence argument into a random process. Draw all variables independently; while some bad event is true, choose one and redraw the variables on which it depends. If there are numbers  $x_A \in (0, 1)$  such that

$$\mathbb{P}(A) \leq x_A \prod_{B \in \Gamma(A)} (1 - x_B) \quad (A \in \mathcal{A}), \quad (15)$$

then the expected number of times  $A$  is redrawn is at most  $x_A/(1 - x_A)$  (Moser and Tardos, 2010). Consequently,

$$\mathbb{E}\tau \leq \sum_{A \in \mathcal{A}} \frac{x_A}{1 - x_A} < \infty. \quad (16)$$

The algorithm therefore stops almost surely and returns an assignment in which no bad event occurs. Bounds of this kind are proved by encoding repair histories as trees, DAGs, or entropy-compression



records whose weights are determined by event probabilities and adjacency in  $D$  (Moser and Tardos, 2010; Haeupler and Harris, 2017; Fialho et al., 2022).

The graph summary omits intersection data. The graph records which repairs may interact, but it does not determine  $\mathbb{P}(A \cap B)$ , set inclusion, or  $\mathbb{P}(\bigcup_A A)$ . Two event systems can therefore have the same graph and the same marginal probabilities but different survival exponents and different nonterminating sets. The next three subsections compare the exact state-space description, the graph bound, and a nested example where the loss is explicit.

### 3.2 The full state space

Every finite variable-model resampling algorithm is a finite-state randomized computation on assignments. Let  $P$  denote the sequential source law of the initial assignment and subsequent resampling draws; finite initial factors do not affect the dimensions below. Delete zero-probability transitions and assume that every surviving transition has probability strictly less than one. Fix a deterministic rule for selecting a currently true bad event. On reachable nonterminal assignments define

$$(B_\theta)_{xy} = \sum_{z: F(x,z)=y} \pi_{v(A(x))}(z)^\theta, \quad (17)$$

where  $A(x)$  is the selected event and  $F(x, z)$  is the updated assignment. Decompose the nonterminal transition graph into reachable strongly connected components  $C$  containing a directed cycle, and let  $B_{\theta,C}$  be the restriction of  $B_\theta$  to  $C$ . Proposition A.1 gives the exact source-relative dimension

$$\dim_H^P \mathcal{N} = \max_C \delta_C, \quad \rho(B_{\delta_C, C}) = 1. \quad (18)$$

This formula is exact, but its matrix usually has exponentially many rows and columns.

### 3.3 The graph bound

Classical entropy-compression and Shearer arguments use the dependency graph and the probabilities  $p_A$  to bound the total weight of possible repair lists (Cartier and Foata, 1969; Scott and Sokal, 2005). Only the complete-graph case is needed here. If  $D = K_m$ , the graph-only enumeration allows every word  $(A_1, \dots, A_t) \in \mathcal{A}^t$  and assigns it weight  $\prod_{i=1}^t p_{A_i}$ . Its total weight is

$$\left( \sum_A p_A \right)^t.$$

This is an upper bound obtained by enumeration from  $D$  and the marginals; it need not equal the probability mass of the histories produced by a fixed selection rule. For uniform  $q$ -ary blocks of size  $b$ , this gives the dimension bound in (29). The bound is classical. A short graph description also does not make exact computation easy: evaluating the relevant independence polynomial and testing membership in the general Shearer region are #P-hard in general (Harvey et al., 2018).

### 3.4 Nested rejection rules

Let  $\Omega$  be a finite block space with source law  $\pi$  satisfying  $0 < \pi(x) < 1$  for every  $x$ , and let  $\mathcal{A}$  be a laminar family of bad subsets: for  $A, B \in \mathcal{A}$ , either  $A \cap B = \emptyset$ ,  $A \subseteq B$ , or  $B \subseteq A$ . Every repair resamples the whole block from  $\pi$ . If the current block  $x$  lies in the bad union  $U = \bigcup_{A \in \mathcal{A}} A$ , select the inclusion-minimal event containing  $x$ . Write  $\phi(x)$  for this event, which is unique by laminarity.

Let  $\text{ch}(A)$  be the maximal proper members of  $\mathcal{A}$  contained in  $A$ , and define the disjoint region

$$C_A = A \setminus \bigcup_{B \in \text{ch}(A)} B. \quad (19)$$

Discard empty regions and let  $\mathcal{I}$  index the remaining ones. They are disjoint, their union is  $U$ , and the algorithm records  $A$  exactly on  $C_A$ . Define

$$w_A(\theta) = \sum_{x \in C_A} \pi(x)^\theta, \quad W(\theta) = \sum_{A \in \mathcal{I}} w_A(\theta) = \sum_{x \in U} \pi(x)^\theta. \quad (20)$$

**Theorem 3.1** (Laminar whole-block rejection). *Assume  $U \neq \emptyset$ , and let  $P = \pi^{\mathbb{N}}$ . There is a unique  $\delta \in [0, 1]$  such that*

$$W(\delta) = \sum_{x \in U} \pi(x)^\delta = 1. \quad (21)$$

(a) *If the alphabet,  $\pi$ ,  $U$ , and the law  $Q_\delta$  below are computable, every nonterminating tape satisfies*

$$\text{Dim}_{\text{eff}, P}(\omega) \leq \delta \quad (\omega \in \mathcal{N} = U^{\mathbb{N}}). \quad (22)$$

*Every tape that is Martin–Löf random for  $Q_\delta^{\mathbb{N}}$  satisfies  $\dim_{\text{eff}, P}(\omega) = \text{Dim}_{\text{eff}, P}(\omega) = \delta$ . Moreover,*

$$\dim_H^P \mathcal{N} = \delta, \quad \mathbb{P}\{\tau \geq t\} = \pi(U)^t. \quad (23)$$

*Under the uniform law on  $Q = q^b$  blocks, writing  $M = |U|$ ,*

$$\dim_H \mathcal{N} = \delta = \frac{\log_2 M}{\log_2 Q}, \quad \mathbb{P}\{\tau \geq t\} = \left(\frac{M}{Q}\right)^t. \quad (24)$$

*In particular, under a uniform source, a tape of effective strong dimension greater than  $\delta$  must make the algorithm terminate.*

(b) *Group rejected blocks according to the region  $C_A$  that contains them. For every  $\theta \geq 0$ , the resulting weighted transition matrix is*

$$\widehat{B}_\theta(A, B) = w_B(\theta). \quad (25)$$

*Its spectral radius is  $W(\theta)$ , the same as that of the full transition matrix, and  $h \equiv 1$  is a right eigenfunction.*

(c) *The dimension and survival bounds are attained by the following reversible record laws. The dimension-attaining law is*

$$Q_\delta(x) = \pi(x)^\delta \mathbf{1}\{x \in U\}. \quad (26)$$

*For every  $T$ -repair execution, there is a probability record law  $G_{\delta, T}$  satisfying*

$$\log_2 \frac{G_{\delta, T}(\text{record})}{P_{\text{tape}}(U_T)} = (1 - \delta) \sum_{t=1}^T \log_2 \frac{1}{\pi(Y_t)}. \quad (27)$$

*The second is the source conditioned on another rejection,  $Q_{\text{surv}}(x) = \pi(x) \mathbf{1}\{x \in U\} / \pi(U)$ , and it satisfies*

$$\log_2 \frac{G_{\text{surv}, T}(\text{record})}{P_{\text{tape}}(U_T)} = T \log_2 \frac{1}{\pi(U)}. \quad (28)$$

*Proof.* The function  $W$  is continuous and strictly decreasing because  $0 < \pi(x) < 1$ . Moreover,  $W(0) = |U| \geq 1$  and  $W(1) = \pi(U) \leq 1$ , so (21) has a unique solution in  $[0, 1]$ .

For full nonterminal states  $x, y \in U$ , resampling the common scope gives  $B_\theta(x, y) = \pi(y)^\theta$ . Hence, for every  $x \in C_A$ ,

$$\sum_{y \in C_B} B_\theta(x, y) = \sum_{y \in C_B} \pi(y)^\theta = w_B(\theta),$$

independently of  $x$ . Thus grouping states by the sets  $C_A$  leaves the weighted transition calculation unchanged and proves (25). The reduced matrix is  $\mathbf{1}w(\theta)^\top$ , so its only nonzero eigenvalue is  $W(\theta)$  and  $h \equiv 1$  is a right eigenfunction. Proposition A.1 now gives  $\dim_H^P \mathcal{N} = \delta$ . Each repair draws a fresh block from  $\pi$  and continues precisely on  $U$ , proving the survival formula and its uniform specialization.

For the dimension bound, observe that  $Q_\delta$  is computable and supported on  $U$ . Every  $\omega \in U^\mathbb{N}$  therefore satisfies

$$K(\omega_{1:n}) \leq -\log_2 Q_\delta^\mathbb{N}[\omega_{1:n}] + O(\log_2 n) = \delta(-\log_2 P[\omega_{1:n}]) + O(\log_2 n).$$

Division by the source information proves (22). If  $\omega$  is Martin–Löf random for  $Q_\delta^\mathbb{N}$ , Levin–Schnorr gives the reverse inequality up to  $O(1)$ , and hence equality. This also proves (5).

It remains to construct the record laws. The repair list is determined by the overwritten states because  $A_t = \phi(Y_t)$ . Reversal recovers the used tape from  $(S_T, Y_{1:T})$ . The source factorization is

$$P_{\text{tape}}(U_T) = \pi(S_T) \prod_{t=1}^T \pi(Y_t).$$

Since (21) holds, the record law

$$G_{\delta,T}(s, y_{1:T}) = \pi(s) \prod_{t=1}^T \left[ \pi(y_t)^\delta \mathbf{1}\{y_t \in U\} \right]$$

has total mass one. Dividing by the source factorization proves (27). Replacing each critical factor by  $\pi(y_t) \mathbf{1}\{y_t \in U\} / \pi(U)$  gives a probability law and proves (28). The region containing  $y_t$  determines the recorded rejection label  $A_t = \phi(y_t)$ .  $\square$

*Remark 3.2* (Beyond laminar families). Laminarity is used only to define the selector cells canonically through set inclusion. For an arbitrary family of whole-block rejection rules with a deterministic selector  $\phi$ , the same conclusions hold with  $C_A = \{x \in U : \phi(x) = A\}$ .

*Remark 3.3* (The one-state reduction). Every repair redraws the whole block. Consequently, the next step is independent of the current rejected block, and the one-state matrix  $[W(\theta)]$  already gives both  $W(1) = \pi(U)$  and the dimension root. The disjoint regions retain the recorded reason for rejection, but do not change the stopping dynamics. Partially overlapping local repairs, which leave relevant state unchanged, are the harder case.

The survival law is also a direct application of Theorem 2.1. Relabel each repair by the disjoint region  $C_A$  containing the overwritten block; this does not change the run or its resampling rule. Put  $r_A = \pi(C_A)$  and choose

$$Q_T(A_1, \dots, A_T) = \prod_{t=1}^T \frac{r_{A_t}}{\pi(U)}.$$

For the relabeled event  $C_{A_t}$ , the event contribution in (8) is  $\log_2(1/r_{A_t})$ , whereas encoding the region label under  $Q_T$  costs  $\log_2(\pi(U)/r_{A_t})$  bits. Their difference is  $\log_2(1/\pi(U))$  at every repair, giving (28). Conditioning only on the original event  $A_t$  would count overlapping mass more than once.

**Comparison with the dependency graph.** Under the standard variable-overlap definition, all common-block events are pairwise adjacent. The graph calculation therefore uses the one-step weight  $\sum_A \pi(A)$ , while the true rejection probability is  $\pi(U)$ . Under a uniform source, the resulting upper bound and the true dimension are

$$1 + \frac{\log_2 \sum_A \pi(A)}{\log_2 Q} \quad \text{and} \quad 1 + \frac{\log_2 \pi(U)}{\log_2 Q}, \quad (29)$$

again truncated to  $[0, 1]$ . They agree when the events are disjoint and differ when events overlap.

In the variable model, a nonedge certifies disjoint scopes and hence commuting repairs; an edge marks only possible interaction. A complete graph therefore contains no information about how the corresponding bad events intersect. The following standard family makes the loss explicit.

**Example 3.4** (Nested Hamming balls). Let  $\Omega = \{0, 1\}^b$  be uniform and reject a block when its Hamming weight is at most  $r$ . Writing

$$U_r = \{x : |x| \leq r\}, \quad M_r = \sum_{j=0}^r \binom{b}{j},$$

Theorem 3.1 gives, for every nonterminating tape,

$$\text{Dim}_{\text{eff}}(\omega) \leq \frac{1}{b} \log_2 M_r \quad \text{for every } \omega \in U_r^{\mathbb{N}}, \quad \dim_H(U_r^{\mathbb{N}}) = \frac{1}{b} \log_2 M_r. \quad (30)$$

Now add the stricter rule  $U_s \subset U_r$ , where  $s < r$ , and record  $U_s$  whenever it applies. The standard variable-overlap dependency graph on the two original rules is  $K_2$  and the graph-only sum is  $(M_s + M_r)/2^b$ , although the computation rejects on a set of mass only  $M_r/2^b$ .

For example, take  $b \geq 3$ ,  $r = b - 1$ , and  $s = b - 2$ . The graph sum is  $2 - (b + 2)/2^b > 1$ , so the graph bound gives only the trivial upper bound 1, whereas the true dimension is  $b^{-1} \log_2(2^b - 1) < 1$ . Splitting  $U_r$  into  $U_s$  and  $U_r \setminus U_s$  removes this double counting.

**Example 3.5** (Same graph and marginals, different dimensions). The graph and bad-event probabilities also fail to identify the exact behavior. Fix  $b \geq 3$  and let

$$B = U_{b-2}, \quad A = \{0^b\}, \quad A' = \{1^b\}.$$

Compare the two event systems  $\{A, B\}$  and  $\{A', B\}$ . Every event depends on all  $b$  block coordinates, so both systems have the same standard variable-overlap dependency graph  $K_2$ . With the singleton event listed first, they also have the same ordered vector of bad-event probabilities:

$$(\pi(A), \pi(B)) = (\pi(A'), \pi(B)) = \left(2^{-b}, \frac{2^b - b - 1}{2^b}\right).$$

Their intersections are different:  $A \subset B$ , while  $A' \cap B = \emptyset$ . Consequently,

$$|A \cup B| = 2^b - b - 1, \quad |A' \cup B| = 2^b - b.$$

After  $t$  redraws, the two survival probabilities are therefore

$$\left(\frac{2^b - b - 1}{2^b}\right)^t \quad \text{and} \quad \left(\frac{2^b - b}{2^b}\right)^t,$$

so their survival exponents are

$$\alpha_1 = b - \log_2(2^b - b - 1), \quad \alpha_2 = b - \log_2(2^b - b).$$

Let  $\mathcal{N}_1 = (A \cup B)^\mathbb{N}$  and  $\mathcal{N}_2 = (A' \cup B)^\mathbb{N}$ . Theorem 3.1 gives

$$\max_{\omega \in \mathcal{N}_1} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_1 = \frac{1}{b} \log_2(2^b - b - 1), \quad \max_{\omega \in \mathcal{N}_2} \text{Dim}_{\text{eff}}(\omega) = \dim_H \mathcal{N}_2 = \frac{1}{b} \log_2(2^b - b).$$

Thus the dependency graph and bad-event probabilities do not determine the exact survival exponent, the maximal effective strong dimension, or the Hausdorff dimension.

This does not contradict tightness results for the Shearer region. Those results concern the worst event system with given graph and marginals; the two systems in Example 3.5 show that fixed systems with those data can behave differently. Related separations in the variable model, based on additional intersection information, appear in He et al. (2023). The present theory is neither an application nor a strengthening of the variable Lovász local lemma (He et al., 2023); rather, for the instances it solves exactly, it quantitatively determines the survival exponent, maximal effective strong dimension, and Hausdorff dimension of nontermination.

### 3.5 Partially overlapping local repairs

The preceding examples are memoryless because every action redraws the same block. A partially overlapping local repair leaves values relevant to later repairs unchanged, and those retained values can affect which event is selected next.

Existing representations of repair histories retain some of this persistent state information (Achlioptas and Iliopoulos, 2016; Haeupler and Harris, 2017; Fialho et al., 2022). A concrete test case is bounded pathwidth. Consider a  $q$ -ary resampling system whose variable–event incidence graph has pathwidth  $w$  and whose deterministic selection rule can be updated from a path-decomposition boundary using  $\text{poly}(n)$  control states. Can the boundary assignment, which contains at most  $w + 1$  active variables, together with this control state define a matrix of size  $q^{O(w)} \text{poly}(n)$  whose spectral radius gives the exact dimension? Can its dimension-attaining change of measure be implemented by reversible execution records, giving matching upper and lower bounds for the effective dimension of a tape? Forbidden-pattern generation and bounded-pathwidth  $k$ -SAT provide two test cases.

A spectral calculation alone gives no code for actual runs. Conversely, a code that counts histories more than once need not give the true dimension.

## 4 Conclusion

This paper studies nontermination at the level of one random tape. Its main theoretical contribution is the backward likelihood identity for finite variable-model resampling computations. The self-information of repaired events minus the code length of the repair sequence is both a change-of-measure quantity controlling long runs and, under computability, a lower bound on the reduction in description length relative to source self-information. The running-time and Kolmogorov-complexity bounds are thus consequences of the same identity.

For rejection sampling and whole-block resampling, the dimension bounds are exact. The law  $Q_\delta$  produces individual tapes attaining the effective dimension bound, while the source-relative Hausdorff dimension measures the entire nonterminating set. Disjoint selector cells recover the exact survival exponent from overlapping rejection rules. Example 3.5 proves that the dependency graph

and bad-event probabilities alone determine neither the survival exponent, the maximal effective strong dimension, nor the Hausdorff dimension.

Partially overlapping local repair with persistent state remains the main unresolved case. A repair label or dependency graph may omit retained values needed to predict and reverse the next step; the full assignment is exact but generally exponential in the number of variables. The open problem is to find a state representation substantially smaller than the full assignment while retaining the information needed to compute the survival exponent and reverse individual runs. Such a representation would permit matching bounds for the survival exponent, individual-tape dimension, and Hausdorff dimension under genuinely local overlap.

## A Finite-state computations

Let  $\mathcal{G}$  be a finite directed multigraph of nonterminal states with fixed initial state  $s_0$ . Each outgoing edge  $e : s \rightarrow t$  represents a distinct next source outcome and has probability  $p(e) > 0$ . The outgoing probabilities from each state sum to at most one; the remaining probability terminates the computation. A surviving path  $w = e_1 \cdots e_n$  determines a tape cylinder of probability

$$P[w] = \prod_{i=1}^n p(e_i).$$

Let  $\mathcal{W}_n$  be the set of surviving paths of length  $n$  from  $s_0$ . Assume that the reachable nonterminal graph is irreducible, contains a directed cycle, and satisfies  $\max_e p(e) < 1$ . For  $\theta \geq 0$ , define

$$(B_\theta)_{st} = \sum_{e:s \rightarrow t} p(e)^\theta, \quad \Lambda(\theta) = \log_2 \rho(B_\theta). \quad (31)$$

The quantity  $\Lambda(\theta)$  is the base-two pressure, the exponential growth rate of the total  $\theta$ -weight of surviving paths. Reducible systems are handled by applying the result to each reachable strongly connected component containing a directed cycle and taking the largest dimension.

Use  $P[w]$  as the diameter of the cylinder  $[w]$ . The resulting Hausdorff dimension is the source-relative dimension  $\dim_H^P$  defined in the introduction; for a uniform  $q$ -ary source it is ordinary Hausdorff dimension in the  $q$ -adic metric.

**Proposition A.1** (The finite-state formula). *There is a unique  $\delta \in [0, 1]$  satisfying  $\rho(B_\delta) = 1$ . Moreover,*

$$\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \sum_{w \in \mathcal{W}_n} P[w]^\theta = \Lambda(\theta), \quad (32)$$

$$\lim_{n \rightarrow \infty} -\frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\} = -\Lambda(1), \quad (33)$$

$$\dim_H^P \mathcal{N} = \delta. \quad (34)$$

If  $B_\delta h = h$  with  $h > 0$ , then

$$Q_\delta(e \mid s) = p(e)^\delta \frac{h_t}{h_s} \quad (35)$$

is a probability law on surviving paths and, for every  $w = e_1 \cdots e_n \in \mathcal{W}_n$ , satisfies

$$\log_2 \frac{Q_\delta[w]}{P[w]} = (1 - \delta) \log_2 \frac{1}{P[w]} + \log_2 \frac{h_{s_n}}{h_{s_0}}. \quad (36)$$

For a computable system with computable  $Q_\delta$ , every nonterminating tape satisfies  $\text{Dim}_{\text{eff}, P}(\omega) \leq \delta$ . A tape that is Martin–Löf random under  $Q_\delta$  satisfies  $\dim_{\text{eff}, P}(\omega) = \text{Dim}_{\text{eff}, P}(\omega) = \delta$ .

*Proof.* Since the graph contains a directed cycle,  $\rho(B_0) \geq 1$ . Since  $B_1$  is substochastic,  $\rho(B_1) \leq 1$ . Irreducibility and  $\max_e p(e) < 1$  make  $\rho(B_\theta)$  continuous and strictly decreasing, proving existence and uniqueness of  $\delta$ . The matrix identity

$$\sum_{w \in \mathcal{W}_n} P[w]^\theta = e_{s_0}^\top B_\theta^n \mathbf{1}$$

and Perron–Frobenius theory prove (32); setting  $\theta = 1$  gives (33).

If  $\theta > \delta$ , then  $\rho(B_\theta) < 1$ , so the total  $\theta$ -weight of the length- $n$  surviving cylinders tends to zero. These cylinders cover  $\mathcal{N}$ , giving  $\dim_H^P \mathcal{N} \leq \delta$ . At  $\theta = \delta$ , (35) has row sums one, and telescoping gives

$$Q_\delta[w] = P[w]^\delta \frac{h_{s_n}}{h_{s_0}}.$$

Because  $h$  is positive on a finite state space, the right side is within a fixed multiplicative factor of  $P[w]^\delta$ . The mass-distribution principle gives  $\dim_H^P \mathcal{N} \geq \delta$ , proving (34) and (36) (Bowen, 1975; Mauldin and Williams, 1988).

For a computable system, coding under  $Q_\delta$  gives, for every surviving prefix  $w$  of length  $n$ ,

$$K(w) \leq -\log_2 Q_\delta[w] + O(\log_2 n) = \delta \log_2 \frac{1}{P[w]} + O(\log_2 n).$$

Division by source self-information proves the upper effective-dimension bound. Levin–Schnorr gives the reverse inequality for a  $Q_\delta$ -Martin–Löf-random path (Lutz, 2003; Athreya et al., 2007; Simpson, 2015).  $\square$

Under a uniform  $q$ -ary source, let  $M_{st}$  count source symbols that move  $s$  to  $t$ . Then  $B_\theta = q^{-\theta} M$ . If

$$\alpha = -\lim_{n \rightarrow \infty} \frac{1}{n} \log_2 \mathbb{P}\{\tau \geq n\}$$

is the survival exponent, then

$$\dim_H \mathcal{N} = \log_q \rho(M) = 1 - \frac{\alpha}{\log_2 q}.$$

Thus, for a uniform finite-state source, the survival exponent determines the dimension. Equation (36) is the finite-state form of the comparison used in Theorem 2.1.

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